

Urban Nexus

Smart City Municipal Management System

Normalization Proofs

Minimal Functional Dependency Sets & BCNF Verification

Table of Contents

1. Citizen	10. Complaint
2. Household [3NF]	11. WorkOrder
3. Property	12. Contractor
4. Ownership [Added]	13. Department
5. UtilityConnection	14. Officer
6. SmartMeter	15. PublicAsset
7. MeterReading	16. MaintenanceSchedule
8. BillingRecord	17. BudgetAllocation
9. PaymentTransaction	18. DepartmentExpense

MINIMAL FD SETS & BCNF PROOFS – ALL 18 RELATIONS

1. Citizen PK: Aadhaar_Number

Relation Schema

Citizen(Aadhaar_Number, first_name, last_name, gender, date_of_birth, mobile_number, email, occupation, registration_date, status)

FD Set (F)

Aadhaar_Number -> first_name
Aadhaar_Number -> last_name
Aadhaar_Number -> gender
Aadhaar_Number -> date_of_birth
Aadhaar_Number -> mobile_number
Aadhaar_Number -> email Aadhaar_Number
-> occupation Aadhaar_Number ->
registration_date Aadhaar_Number -> status

Minimal FD Set (Fc) Aadhaar_Number

-> first_name Aadhaar_Number ->
last_name Aadhaar_Number ->
gender
Aadhaar_Number -> date_of_birth
Aadhaar_Number -> mobile_number
Aadhaar_Number -> email Aadhaar_Number
-> occupation Aadhaar_Number ->
registration_date Aadhaar_Number -> status

Attribute Closure

Aadhaar_Number+ = {Aadhaar_Number, first_name, last_name, gender, date_of_birth, mobile_number, email, occupation, registration_date, status}

Candidate Key: Aadhaar_Number BCNF Proof

The only non-trivial FDs have Aadhaar_Number as the left-hand side. Since Aadhaar_Number is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD $X \rightarrow Y$, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

2. Household PK: household_id

Household Relation

Household(household_id, head_aadhaar, total_members, ward_number, address_line1, address_line2, pincode)

Location Relation

Location(pincode, city, state)

Functional Dependencies

Household:

$\text{household_id} \rightarrow \{\text{head_aadhaar}, \text{total_members}, \text{ward_number}, \text{address_line1}, \text{address_line2}, \text{pincode}\}$

Location:

$\text{pincode} \rightarrow \{\text{city}, \text{state}\}$

Minimal Functional Dependencies

Household:

$\text{household_id} \rightarrow \{\text{head_aadhaar}, \text{total_members}, \text{ward_number}, \text{address_line1}, \text{address_line2}, \text{pincode}\}$

Location:

$\text{pincode} \rightarrow \{\text{city}, \text{state}\}$

Attribute Closure

$\text{household_id}^+ = \{\text{household_id}, \text{head_aadhaar}, \text{total_members}, \text{ward_number}, \text{address_line1}, \text{address_line2}, \text{pincode}\}$

$\text{pincode}^+ = \{\text{pincode}, \text{city}, \text{state}\}$

Candidate Keys

Household: household_id

Location: pincode

BCNF Proof

Both relations are in BCNF because for every non-trivial functional dependency, the left-hand side is a superkey.

3. Property PK: property_id

Relation Schema

Property(property_id, household_id, property_type, property_use_type, built_up_area_sqft, land_area_sqft, construction_year, zone, ward_number, property_status)

FD Set (F)

property_id -> household_id
property_id -> property_type
property_id -> property_use_type
property_id -> built_up_area_sqft
property_id -> land_area_sqft
property_id -> construction_year
property_id -> zone
property_id -> ward_number property_id ->
property_status

Minimal FD Set (Fc) property_id

-> household_id property_id ->
property_type
property_id -> property_use_type
property_id -> built_up_area_sqft
property_id -> land_area_sqft property_id
-> construction_year property_id -> zone
property_id -> ward_number property_id ->
property_status

Attribute Closure

property_id⁺ = {property_id, household_id, property_type, property_use_type, built_up_area_sqft, land_area_sqft, construction_year, zone, ward_number, property_status}

Candidate Key: property_id

BCNF Proof

The only non-trivial FDs have property_id as the left-hand side. Since property_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

4. Ownership PK: ownership_id

Relation Schema

Ownership(ownership_id, property_id, aadhaar_number, ownership_percentage, ownership_start_date, ownership_end_date)

FD Set (F)

ownership_id $\rightarrow$ property_id
ownership_id $\rightarrow$ aadhaar_number
ownership_id $\rightarrow$ ownership_percentage
ownership_id $\rightarrow$ ownership_start_date
ownership_id $\rightarrow$ ownership_end_date

Minimal FD Set (Fc) ownership_id $\rightarrow$

property_id
aadhaar_number
ownership_percentage
ownership_start_date
ownership_end_date

Attribute Closure

ownership_id⁺ = {ownership_id, property_id, aadhaar_number, ownership_percentage, ownership_start_date, ownership_end_date}

Candidate Key: ownership_id BCNF

Proof

The only non-trivial FDs have ownership_id as the left-hand side. Since ownership_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey $\rightarrow$ attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD $X \rightarrow Y$, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

5. UtilityConnection PK: connection_id

Relation Schema

UtilityConnection(connection_id, property_id, utility_type, connection_date, connection_number, tariff_category, connection_status)

FD Set (F)

connection_id -> property_id connection_id
-> utility_type connection_id ->
connection_date connection_id ->
connection_number connection_id ->
tariff_category connection_id ->
connection_status

Minimal FD Set (Fc) connection_id ->

property_id connection_id ->
utility_type connection_id ->
connection_date
connection_id -> connection_number
connection_id -> tariff_category
connection_id -> connection_status

Attribute Closure

connection_id⁺ = {connection_id, property_id, utility_type, connection_date, connection_number, tariff_category, connection_status}

Candidate Key: connection_id BCNF

Proof

The only non-trivial FDs have connection_id as the left-hand side. Since connection_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD $X \rightarrow Y$, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

6. SmartMeter PK: meter_id

Relation Schema

SmartMeter(meter_id, connection_id, manufacturer, meter_serial_number, installation_date, meter_status)

FD Set (F)

meter_id -> connection_id
meter_id -> manufacturer
meter_id -> meter_serial_number
meter_id -> installation_date
meter_id -> meter_status

Minimal FD Set (Fc) meter_id ->

connection_id
meter_id -> meter_serial_number
meter_id -> installation_date
meter_id -> meter_status

Attribute Closure

meter_id⁺ = {meter_id, connection_id, manufacturer, meter_serial_number, installation_date, meter_status}

Candidate Key: meter_id BCNF

Proof

The only non-trivial FDs have meter_id as the left-hand side. Since meter_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

7. MeterReading PK: reading_id

Relation Schema

MeterReading(reading_id, meter_id, reading_date, reading_value, reading_unit, recorded_by, remarks)

FD Set (F)

reading_id -> meter_id
reading_id -> reading_date
reading_id -> reading_value
reading_id -> reading_unit
reading_id -> recorded_by
reading_id -> remarks

Minimal FD Set (Fc) reading_id ->

meter_id
reading_date
reading_value
reading_id -> reading_unit
reading_id -> recorded_by
reading_id -> remarks

Attribute Closure

reading_id⁺ = {reading_id, meter_id, reading_date, reading_value, reading_unit, recorded_by, remarks}

Candidate Key: reading_id BCNF

Proof

The only non-trivial FDs have reading_id as the left-hand side. Since reading_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

8. BillingRecord PK: bill_id

Relation Schema

BillingRecord(bill_id, connection_id, billing_period_start, billing_period_end, previous_reading, current_reading, units_consumed, bill_amount, due_date, bill_status, late_fee, generated_date)

FD Set (F)

bill_id -> connection_id
bill_id -> billing_period_start
bill_id -> billing_period_end
bill_id -> previous_reading
bill_id -> current_reading
bill_id -> units_consumed
bill_id -> bill_amount
bill_id -> due_date
bill_id -> bill_status
bill_id -> late_fee
bill_id -> generated_date

Minimal FD Set (Fc)

bill_id -> connection_id
bill_id -> billing_period_start
bill_id -> billing_period_end
bill_id -> previous_reading
bill_id -> current_reading
bill_id -> units_consumed
bill_id -> bill_amount
bill_id -> due_date
bill_id -> bill_status
bill_id -> late_fee
bill_id -> generated_date

Attribute Closure

bill_id⁺ = {bill_id, connection_id, billing_period_start, billing_period_end, previous_reading, current_reading, units_consumed, bill_amount, due_date, bill_status, late_fee, generated_date}

Candidate Key: bill_id

BCNF Proof

The only non-trivial FDs have bill_id as the left-hand side. Since bill_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

9. PaymentTransaction PK: payment_id

Relation Schema

PaymentTransaction(payment_id, bill_id, payment_date, payment_amount, payment_mode, transaction_reference, payment_status)

FD Set (F)

payment_id -> bill_id payment_id ->
payment_date payment_id ->
payment_amount payment_id ->
payment_mode
payment_id -> transaction_reference
payment_id -> payment_status

Minimal FD Set (Fc) payment_id ->

bill_id payment_id -> payment_date
payment_id -> payment_amount
payment_id -> payment_mode payment_id
-> transaction_reference payment_id ->
payment_status

Attribute Closure

payment_id⁺ = {payment_id, bill_id, payment_date, payment_amount, payment_mode, transaction_reference, payment_status}

Candidate Key: payment_id BCNF

Proof

The only non-trivial FDs have payment_id as the left-hand side. Since payment_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

10. Complaint PK: complaint_id

Relation Schema

Complaint(complaint_id, aadhaar_number, property_id, complaint_category, complaint_description, complaint_date, priority_level, status, assigned_department_id)

FD Set (F)

complaint_id $\rightarrow$ aadhaar_number
complaint_id $\rightarrow$ property_id complaint_id $\rightarrow$
complaint_category complaint_id $\rightarrow$
complaint_description complaint_id $\rightarrow$
complaint_date complaint_id $\rightarrow$
priority_level complaint_id $\rightarrow$ status
complaint_id $\rightarrow$ assigned_department_id

Minimal FD Set (Fc) complaint_id $\rightarrow$

aadhaar_number complaint_id $\rightarrow$
property_id
complaint_id $\rightarrow$ complaint_category
complaint_id $\rightarrow$ complaint_description
complaint_id $\rightarrow$ complaint_date
complaint_id $\rightarrow$ priority_level complaint_id
 $\rightarrow$ status
complaint_id $\rightarrow$ assigned_department_id

Attribute Closure

complaint_id⁺ = {complaint_id, aadhaar_number, property_id, complaint_category, complaint_description, complaint_date, priority_level, status, assigned_department_id}

Candidate Key: complaint_id BCNF

Proof

The only non-trivial FDs have complaint_id as the left-hand side. Since complaint_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey $\rightarrow$ attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD $X \rightarrow Y$, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

11. WorkOrder PK: work_order_id

Relation Schema

WorkOrder(work_order_id, complaint_id, assigned_officer_id, contractor_id, issue_date, expected_completion_date, actual_completion_date, work_status, cost_incurred)

FD Set (F)

work_order_id -> complaint_id
work_order_id -> assigned_officer_id
work_order_id -> contractor_id
work_order_id -> issue_date
work_order_id -> expected_completion_date work_order_id ->
actual_completion_date work_order_id -> work_status
work_order_id -> cost_incurred

Minimal FD Set (Fc)

work_order_id -> complaint_id
work_order_id -> assigned_officer_id
work_order_id -> contractor_id
work_order_id -> issue_date
work_order_id -> expected_completion_date work_order_id ->
actual_completion_date work_order_id -> work_status
work_order_id -> cost_incurred

Attribute Closure

work_order_id⁺ = {work_order_id, complaint_id, assigned_officer_id, contractor_id, issue_date, expected_completion_date, actual_completion_date, work_status, cost_incurred}

Candidate Key: work_order_id BCNF

Proof

The only non-trivial FDs have work_order_id as the left-hand side. Since work_order_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

12. Contractor PK: contractor_id

Relation Schema

Contractor(contractor_id, contractor_name, contractor_type, email, address, license_number, registration_date, contact_person, mobile_number, status)

FD Set (F)

contractor_id -> contractor_name
contractor_id -> contractor_type
contractor_id -> email contractor_id
-> address contractor_id ->
license_number contractor_id ->
registration_date contractor_id ->
contact_person contractor_id ->
mobile_number contractor_id ->
status

Minimal FD Set (Fc) contractor_id ->

contractor_name contractor_id ->
contractor_type contractor_id ->
email
contractor_id -> address contractor_id ->
license_number contractor_id ->
registration_date contractor_id ->
contact_person contractor_id ->
mobile_number contractor_id -> status

Attribute Closure

contractor_id⁺ = {contractor_id, contractor_name, contractor_type, email, address, license_number, registration_date, contact_person, mobile_number, status}

Candidate Key: contractor_id BCNF

Proof

The only non-trivial FDs have contractor_id as the left-hand side. Since contractor_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD $X \rightarrow Y$, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

13. Department PK: department_id

Relation Schema

Department(department_id, department_name, department_head_id, contact_number, office_location, creation_date)

FD Set (F)

department_id -> department_name
department_id -> department_head_id
department_id -> contact_number
department_id -> office_location
department_id -> creation_date

Minimal FD Set (Fc)

department_id -> department_name
department_id -> department_head_id
department_id -> contact_number
department_id -> office_location
department_id -> creation_date

Attribute Closure

department_id⁺ = {department_id, department_name, department_head_id, contact_number, office_location, creation_date}

Candidate Key: department_id BCNF

Proof

The only non-trivial FDs have department_id as the left-hand side. Since department_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

14. Officer PK: officer_id

Relation Schema

Officer(officer_id, first_name, last_name, designation, department_id, mobile_number, email, joining_date, employment_status)

FD Set (F)

officer_id -> first_name officer_id ->
last_name officer_id -> designation
officer_id -> department_id officer_id
-> mobile_number officer_id -> email
officer_id -> joining_date officer_id
-> employment_status

Minimal FD Set (Fc) officer_id ->

first_name officer_id ->
last_name officer_id ->
designation
officer_id -> department_id officer_id
-> mobile_number officer_id -> email
officer_id -> joining_date officer_id
-> employment_status

Attribute Closure

officer_id+ = {officer_id, first_name, last_name, designation, department_id, mobile_number, email, joining_date, employment_status}

Candidate Key: officer_id BCNF

Proof

The only non-trivial FDs have officer_id as the left-hand side. Since officer_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

15. PublicAsset PK: asset_id

Relation Schema

PublicAsset(asset_id, asset_name, asset_type, location, ward_number, installation_date, asset_condition, responsible_department_id)

FD Set (F)

asset_id -> asset_name asset_id
-> asset_type asset_id ->
location asset_id ->
ward_number asset_id ->
installation_date asset_id ->
asset_condition
asset_id -> responsible_department_id

Minimal FD Set (Fc) asset_id ->

asset_name asset_id ->
asset_type asset_id -> location
asset_id -> ward_number asset_id ->
installation_date asset_id ->
asset_condition
asset_id -> responsible_department_id

Attribute Closure

asset_id+ = {asset_id, asset_name, asset_type, location, ward_number, installation_date, asset_condition, responsible_department_id}

Candidate Key: asset_id BCNF

Proof

The only non-trivial FDs have asset_id as the left-hand side. Since asset_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

16. MaintenanceSchedule PK: schedule_id

Relation Schema

MaintenanceSchedule(schedule_id, asset_id, scheduled_date, assigned_officer_id, maintenance_type, status)

FD Set (F)

schedule_id -> asset_id
schedule_id -> scheduled_date
schedule_id -> assigned_officer_id
schedule_id -> maintenance_type
schedule_id -> status

Minimal FD Set (Fc) schedule_id ->

asset_id
scheduled_date
assigned_officer_id
maintenance_type
status

Attribute Closure

schedule_id⁺ = {schedule_id, asset_id, scheduled_date, assigned_officer_id, maintenance_type, status}

Candidate Key: schedule_id BCNF

Proof

The only non-trivial FDs have schedule_id as the left-hand side. Since schedule_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition. **[BCNF] This relation is in BCNF.**

17. BudgetAllocation PK: budget_id

Relation Schema

BudgetAllocation(budget_id, department_id, financial_year, allocated_amount, approved_by, approved_date)

FD Set (F)

budget_id -> department_id
budget_id -> financial_year budget_id
-> allocated_amount budget_id ->
approved_by budget_id ->
approved_date

Minimal FD Set (Fc) budget_id ->

department_id budget_id ->
financial_year
budget_id -> allocated_amount
budget_id -> approved_by budget_id
-> approved_date

Attribute Closure

budget_id+ = {budget_id, department_id, financial_year, allocated_amount, approved_by, approved_date}

Candidate Key: budget_id BCNF

Proof

The only non-trivial FDs have budget_id as the left-hand side. Since budget_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD X -> Y, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.

18. DepartmentExpense PK: expense_id

Relation Schema

DepartmentExpense(expense_id, department_id, budget_id, expense_amount, expense_category, description, expense_date)

FD Set (F)

expense_id -> department_id
expense_id -> budget_id expense_id
-> expense_amount expense_id ->
expense_category expense_id ->
description expense_id ->
expense_date

Minimal FD Set (Fc) expense_id ->

department_id expense_id ->
budget_id expense_id ->
expense_amount
expense_id -> expense_category
expense_id -> description
expense_id -> expense_date

Attribute Closure

expense_id+ = {expense_id, department_id, budget_id, expense_amount, expense_category, description, expense_date}

Candidate Key: expense_id BCNF

Proof

The only non-trivial FDs have expense_id as the left-hand side. Since expense_id is the sole candidate key (and therefore a superkey), every FD in Fc is of the form: superkey -> attribute. By definition, a relation is in BCNF if and only if for every non-trivial FD $X \rightarrow Y$, X is a superkey. All FDs here satisfy this condition.

[BCNF] This relation is in BCNF.